St. George's University
Grenada, West Indies

Basic Principles of Medicine II

Module: Nervous System and Behavioral Sciences | Lecture NB 25

Flipped Class: Clinical Cases and Imaging of Cranial Nerves III, V and VII

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Your Objectives will show here!

Objectives

SOM.MKII.BPM2.1.NB.1.ANAT.0242	Describe the cause of cavernous sinus syndrome, the nerves that may be affected and what consequences can occur
SOM.MKII.BPM2.1.NB.1.ANAT.0244	Describe the intracranial position of the cranial nerves and relate this to surrounding blood vessels. Include in your discussion, consequences of cranial nerve impingement due to aneurysms in the vessels of the cranial circulation
SOM.MKII.BPM2.1.NB.1.ANAT.0245	Describe the causes and locations of intracranial hematomas including epidural hematoma, subdural hematoma and subarachnoid hemorrhage
SOM.MKII.BPM2.1.NB.1.ANAT.0247	List the nerves and muscles that may be affected by the loss of accommodation
SOM.MKII.BPM2.1.NB.1.ANAT.0248	List the nerves and muscles involved in ptosis and partial ptosis of the eyelid
SOM.MKII.BPM2.1.NB.1.ANAT.0249	Describe the signs and symptoms of facial paralysis, Bell's palsy, Horner's syndrome, trigeminal neuralgia
SOM.MKII.BPM2.1.NB.1.ANAT.0250	Describe the signs and symptoms of (shingles) herpes zoster
SOM.MKII.BPM2.1.NB.1.ANAT.0251	Describe the consequence of intracranial lesions of each of the 12 cranial nerves, their nuclei and their ganglion (review from previous lectures)

Recommended readings

Drake RL, Vogl AW, Mitchell AWM. Gray's anatomy for students. 4th ed. Elsevier. Chapter 8.

Siegel and Sapru Essential Neuroscience 4th ed. Chapter 13.

Neuroanatomy in clinical context. An atlas of structures, sections, systems and syndromes. Haines DE. 10th ed. Wolters Kluwer. Chapter 3

Pay special attention to the green 'In the Clinic' boxes which contain the clinical correlates for this topic.

Case 1

A 17-year-old girl is brought to the emergency department by her mother because of 1-hour history of severe headache, swelling and redness around the right eye.

- Her vitals: temperature 39°C (102.2 °F), pulse 86/min, blood pressure 105/75 mm Hg, respiration 17/min.
- Physical examination: Right periorbital edema with bulging of the eye, lid erythema, chemosis, ptosis, proptosis and loss of sensation of the upper and middle of her right face.

What is the likely anatomical location affected?

Which nerves are involved ?

What should the initial investigation include?

Case 2

An 81-year-old woman comes to the physician because of 4-day history of a pain, burning, and numbness of the right side of face. She describes a stinging, tingling, aching sensation.

Physical examination shows vesicular rash, forming small blisters filled with a serous exudate. Diagnosis of herpes zoster is made involving maxillary and ophthalmic branches of the trigeminal nerve.

Where does the virus stay dormant before the development of this condition?

Case 3

A 13-year-old boy is brought to the physician by his mother due to a 3-week history of persistent nosebleeds.

The symptoms began after he was accidentally struck on the nose during a football game. He reports that the bleeding initially started as a single, heavy episode immediately following the impact but has since recurred almost daily, particularly when he engages in physical activity or touches his nose.

Physical examination reveals a hyperemic area on the left anterior nasal septum, an area of anastomosis of blood vessels where the bleeding typically originates.

What are the blood vessels involved?

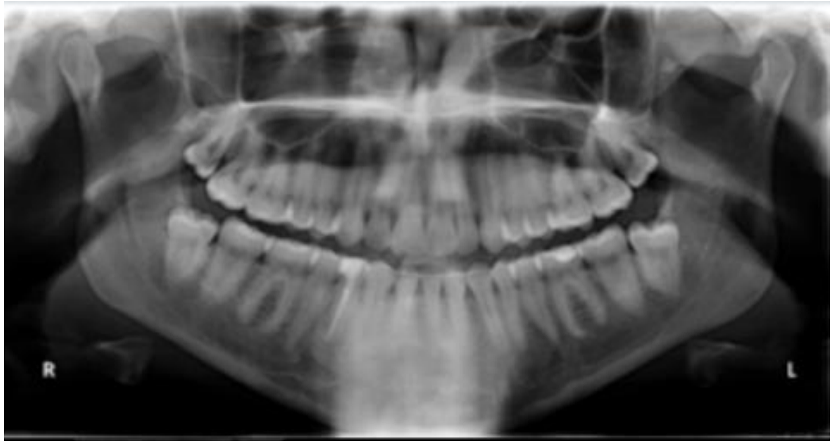
Case 4

A 35-year-old man came to the dentist for right wisdom tooth extraction. CBCT scan shows close or intimate relationship of the mandibular canal with the third mandibular molar root.

One day after the surgery he complained of abnormal sensations in the right chin, lower teeth, lower jaw and lower lips for the last 2 days. He had right wisdom tooth removal 3 days ago.

- Physical examinations shows diminished sensation on the right lower chin area, with normal speech and chewing.

Which nerve is most likely damaged?



Case 5

A 53-year-old woman is brought by her husband to the emergency department because of pain in the back of the right eye for the last 6 hours. She has a one-day history of seeing some colorful and glaring waves in the upper visual field of the right eye lasting for seconds.

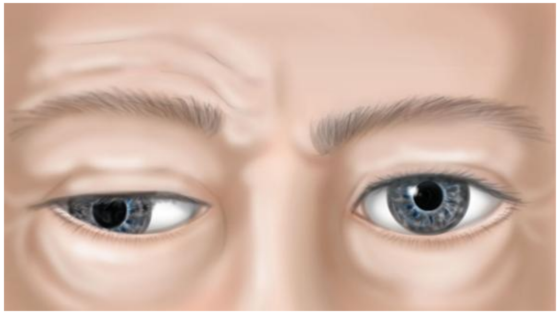
Her vital signs: temperature 38°C (100.4 °F), pulse 65/min, blood pressure 110/75 mm Hg, respiration 23/min. Physical examination shows the right eye was displaced outward "exotropia" and downward "hypotropia“, with enlarged unreactive pupil.

What nerve is affected?

What artery is involved?

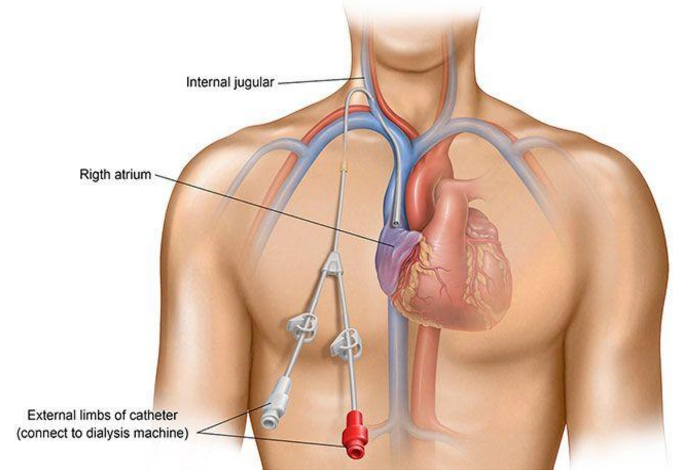
What should the initial investigations include?

What is a likely complication?



Case 6

A 25-year-old female is diagnosed with end-stage renal disease and placed on dialysis. A left internal jugular permacath was inserted. The procedure was uneventful.



2 days later, the patient developed swelling of the left eye

- **Examination** shows drooping upper eyelid; contracted pupil; dryness (lack of sweating) on the same side of the face.

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